

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (EC) Examination

April 2013

EC210 Analog Communication

Date: 26.04.2013, Friday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) Describe various properties of Fourier transform in detail. [07]
- Q - 2 (a) Explain block diagram of Amplitude Modulator transmitter in detail. [07]
- (b) An amplitude modulator transmitter radiates 50w with carrier unmodulated, 50% of modulation index. Calculate total power radiated by amplitude modulated transmitter. And also find same for case of
- 1) 100% modulation index
 - 2) 70% modulation index

OR

- Q - 2 (a) Explain the detection of Amplitude Modulated signal using diode detector. [07]
- (b) How modulation index (depth), effective voltage and effective current are calculated in nonsinusoidal modulation? Derive its equation. [07]
- Q - 3 (a) What are the different types of single sideband generation techniques? Describe balance modulator filter method with upconversion concept. [04]
- (b) A 500w carrier is modulated on modulation index of 100% ,calculate the total power in modulated wave in following form [05]
- 1) Amplitude Modulation(AM)
 - 2) Double sideband suppressed carrier(DSBSC)
 - 3) Single sideband (SSB)
- (c) Define modulation index and deviation ratio in frequency modulation. [05]

OR

- Q - 3 (a) Describe phase shift method of single sideband modulation system. [04]
- (b) Compare single sideband system with amplitude modulation in detail. [05]
- (c) Explain frequency modulation using varactor diode method. [05]

SECTION – II

- Q - 4 (a) What is preemphasis and deemphasis? Explain preemphasis and deemphasis circuit in detail with its need. [07]
- Q - 5 (a) What are the different techniques of frequency demodulation? Explain Ratio detector with its circuit diagram. [06]
- (b) What is Super Heterodyne principle? Draw and explain Super Heterodyne Receiver with each block functionality. [08]

OR

- Q - 5 (a) Describe Quadrature detector for frequency demodulation. [06]
- (b) Explain gain and sensitivity of receiver in detail. [08]
- Q - 6 (a) What is image rejection and double spotting? [03]
- (b) 1. The Noise figure of an amplifier is 7 dB. Calculate the output signal to noise ratio when the input signal to noise ratio is 35 dB. [03]
2. Define Noise Figure. [01]
3. Burst noise is also called as a _____ noise. [01]
- (c) Describe following all terms. [06]
1. Thermal Noise
 2. Shot Noise

OR

- Q - 6 (a) Define following all terms. [03]
- 1) Simple Automatic gain control
 - 2) Delayed Automatic gain control
- (b) A Mixer stage has a noise figure of 20 dB, and this is preceded by an amplifier that has a noise figure of 9 dB and available power gain of 15 dB. Calculate the overall noise figure referred to the input. [05]
- (c) Derive the Friis's formula for N number of amplifier connected in cascade connection. [06]
